

Optimizing Absolute Binding Free Energy Calculations for Production Usage

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Abstract

Protein-ligand binding affinity prediction is a key aspect in computational small molecule drug discovery. Several recent studies have demonstrated that molecular simulations based on alchemical *absolute* binding free-energy (ABFE) calculations are an accurate and broadly applicable tool for this purpose. However, the use of current ABFE protocols in large scale drug discovery projects occasionally leads to unstable simulations and poor convergence. To address these problems, we have implemented several optimizations of the ABFE protocol. First, a new algorithm to choose the protein-ligand pose restraints was developed to prevent numerical instabilities. By considering protein-ligand hydrogen bonds, the restraint selection now incorporates data on the key interactions to improve the convergence. Second, an optimization of the annihilation protocol was conducted to minimize the resulting error of the free energy. Third, a rearrangement of the order with which interactions (electrostatics, Lennard-Jones, restraints, intramolecular torsions) are scaled resulted in a systematic improvement of the precision. The results from four protein-ligand benchmark systems (TYK2, P38, JNK1, and CDK2) show significantly lower variances of the free energy results and

improvements of up to 0.23 kcal/mol for the root mean square error, compared to the original protocol.

Introduction

The ability to predict potency with reliable accuracy is an important capability for small molecule drug discovery projects. Generally, the list of compounds that offer potential improvements of potency is much larger than what could be tested with chemical synthesis. Hence, there is a large variety of tools to achieve potency prediction within rapid chemistry Design-Make-Test-Learn (DMTL) cycles or large-scale activity screens.^{1,2} In general, techniques that rely only on the 2D compound structure are computationally inexpensive but struggle with issues like prediction of activity cliffs or the quality of novel interactions.³ Techniques that use 3D structural information are better able to predict activity cliffs and novel interactions, and methods that additionally use dynamical information offer more rigorous predictions across chemical space.⁴ However, there is a difficult balance to strike between cost/speed, accuracy/precision and robustness/ease-of-application. 3D dynamical potency prediction techniques are more costly to setup and carry out, although no systematic study has focused on these comparisons. In addition, protocols found to perform well in benchmark comparisons may turn out to be not optimal for real-world applications.^{5,6}

One of the most popular 3D dynamical potency prediction techniques are relative binding free-energy (RBFE) calculations, which have been used in real world, structurally enabled small molecule drug discovery projects for many years.⁷ RBFE calculations interconvert one ligand to another along an alchemical reaction coordinate λ and can be used to compare the relative potency of the two involved ligands. However, RBFE calculations are not able to predict the binding affinity for a particular ligand. To overcome this problem, alchemical *absolute* binding free-energy (ABFE) calculations have become more widely applied recently, with various protocols in use within drug discovery companies.⁸⁻¹⁰ ABFE calculations rely on the use of annihilation protocols, which convert the physical ligand to a noninteracting ideal gas state (dummy molecule). In contrast to RBFE, ABFE does not require a complex network of alchemical transformations, which makes it more user friendly and generally applicable.

The use of both RBFE and ABFE protocols in large scale drug discovery projects often reveals a series of challenging edge-cases, which arise from the exploration of a vast diversity of binding pockets and ligands. Problems are often related to the force field, system setup, and convergence/sampling.¹¹⁻¹³ These problems are especially acute in ABFE calculations, which require the complete removal of all interactions between the compound and receptor. Therefore, ABFE requires more extensive sampling, and the convergence of ABFE heavily relies on the application of restraints that keep the compound in the binding pocket.¹⁴ To maximize efficiency, the restraints should maintain the same binding pose for the physical

ligand and the noninteracting dummy state, which increases the phase space overlap during the transformation. Afterwards, the careful removal of the bias due to the imposed restraints is needed to ensure accurate free energy results.¹²

Our initial attempt to build a robust and efficient ABFE workflow was based on a protocol that was initially described by Aldeghi et al., and which has been refined over the years.^{4,8,15} We implemented this protocol in the BioSimSpace bio-simulation framework.¹⁶ However, tests on established benchmark sets,^{9,17} as well as in-house applications, revealed several shortcomings. Occasionally, simulations crashed. We traced this to instantaneous values of the restraint angles approaching 0 or 180 degrees, causing the calculation of the gradient to become numerically unstable (see also Theory & Methods).¹³ In Figure S1, we show the time series of the restraint degrees of freedom for a crashed simulation of an in-house compound. We therefore, developed an automated restraint selection procedure which avoided such instabilities as much as possible.

Moreover, the Aldeghi protocol uses a relatively small number of λ -states, but extensive sampling of 10ns¹⁵ or 20ns⁴ at each state. Inspired by Chen et al.,⁹ we explored protocols with more λ -states but shorter simulation lengths per λ -state to reduce the cost and wall-time of our ABFE workflow. Instead of using equidistant intermediate states (linear or semi-linear λ -scheduling), we searched for an optimized λ -scheduling for turning off nonbonded interactions, in particular the Lennard-Jones (LJ) interactions. Additional analyses of the numerical instabilities due to restraints suggested to modify the order in which interactions and restraints are scaled. Finally, the automated search for restraints and choice of starting coordinates in the Aldeghi protocol¹⁵ lead to rather high variability of binding free energies in independently repeated calculations. Therefore, another goal for the restraint generation algorithm and the λ -scheduling was to increase reproducibility and precision of the results.

In Theory & Methods, we investigate each of the above-mentioned issues in more detail and fully describe the differences between our initial Aldeghi-like protocol and our final protocol. These will be referred to as “Aldeghi” and “RXXR”, respectively. In Results, we first report findings relevant to the individual optimizations, some of which were crucial to the protocol modifications made. We then present results for the TYK2, P38, JNK1, and CDK2 protein-ligand benchmark datasets¹⁷ and compare the results obtained with the RXXR to the Aldeghi ABFE protocol, as well as to the RBFE and ABFE results reported in a study by Chen et al.⁹

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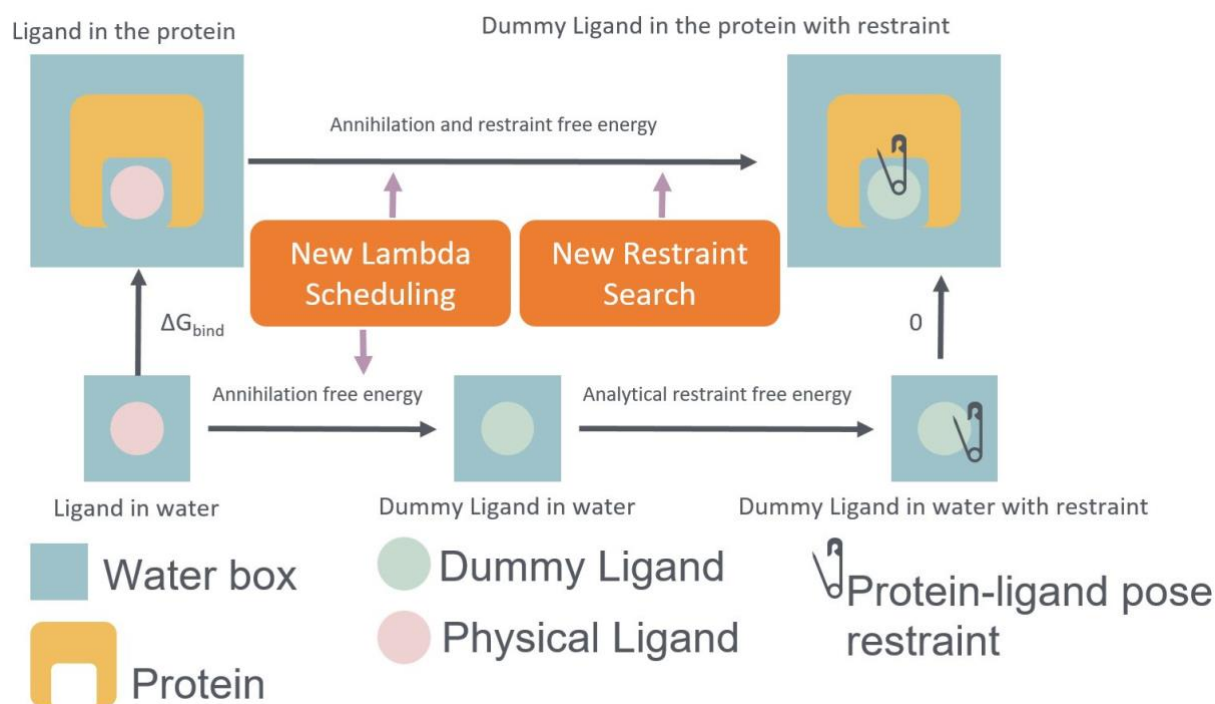


Figure 1. The free energy cycle for calculating the absolute binding free energy (ABFE). The binding free energy ΔG_{bind} of a compound is determined by the difference in free energy between the ligand in water (bottom) and the ligand in the binding pocket (top). Instead of calculating this difference directly, the process is broken down into a series of steps that involve the alchemical annihilation of the physical ligand to a noninteracting dummy ligand. This involves a) the annihilation of the ligand in water to determine the desolvation free energy; b) the free energy associated with applying restraints to the dummy ligand in water; c) the transfer of the dummy ligand into the binding pocket, and d) the free energy required to remove the restraints and create the physical ligand within the binding pocket. The novel methods (orange) are employed in stages a and d for deriving restraints and scheduling the lambda windows.

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The binding free energy of a compound to a target protein is the free energy difference between the ligand in solution and the ligand within the protein's binding pocket. However, calculating this free energy difference directly is computationally demanding. One way¹⁸ to address this is the construction of a thermodynamic cycle¹⁹ as shown in Figure 1. In this cycle, the physical ligand is alchemically transformed into a "dummy" ligand, which lacks electrostatic and LJ interactions. This annihilation is performed both in the binding pocket and in aqueous solution, and their free energy differences are subtracted. To aid convergence, a set of restraints as suggested by Boresch et al. (3 dihedral (ϕ_A , ϕ_B and ϕ_C), 2 angle (θ_A and θ_B) and 1 distance (r) restraint)²⁰ is applied when the ligand is annihilated in the binding pocket, with the associated free energy contribution being calculated numerically. The free

energy of applying these restraints to the dummy ligand in the solvent is determined analytically.²¹

Implementation of the two ABFE protocols

The physics-based drug discovery platform (henceforth referred to as physics platform) in use at Recursion is built upon BioSimSpace¹⁶ and implemented in ArgoWorkflows,²² enabling fully automated and scalable calculations using cloud resources. The two ABFE workflows compared in this study were both implemented in this framework (BioSimSpace¹⁶ /ArgoWorkflows²², see Figure S2). The initial protocol was a reimplement of a workflow introduced by Aldeghi et al., which is widely used.^{4,15} This workflow is briefly outlined below^{4,15}, followed by a description of the substantial modifications and optimizations that led to the new RXXR protocol.

Restraint search for the two ABFE protocols

A key step of the alchemical ABFE thermodynamic cycle (Figure 1) is the careful selection of restraints to keep the ligand in a position and orientation that resembles the bound state. When using the six restraints suggested by Boresch et al.,²⁰ numerical instabilities can arise if the instantaneous values of the restraint angles θ_A and/or θ_B become 0 or 180 degrees.¹³ This poses a significant challenge to the fully automated selection of the restraint degrees of freedom.

The Aldeghi protocol employs the MDRestraintsGenerator⁴ tool for generating restraints. The MDRestraintsGenerator selects the least mobile ligand atoms from the 1 ns equilibration phase as candidate anchor points for orientational restraints. Protein anchor points are chosen from C α -carbons within an 8 Å cutoff from the ligand anchors. A set of potential orientational restraints is then generated by selecting the closest ligand heavy atoms and protein backbone atoms. Restraint candidates with angles greater than 155 degrees or less than 25 degrees are excluded. For each remaining candidate, bond, angle, and dihedral timeseries are analysed, and the restraint with the lowest standard deviation across all values is selected. Since the initial ligand atom is chosen based on RMSF (Root Mean Square Fluctuation), generating restraints from different equilibration trajectories can yield varying results. In our hands, restraints generated in this manner occasionally resulted in unstable simulations. While rerunning the equilibration and generating a new restraint resolves the issue in most cases, such infrequent instabilities are undesirable for a robust automated pipeline, because they increase the costs and delay drug design projects.

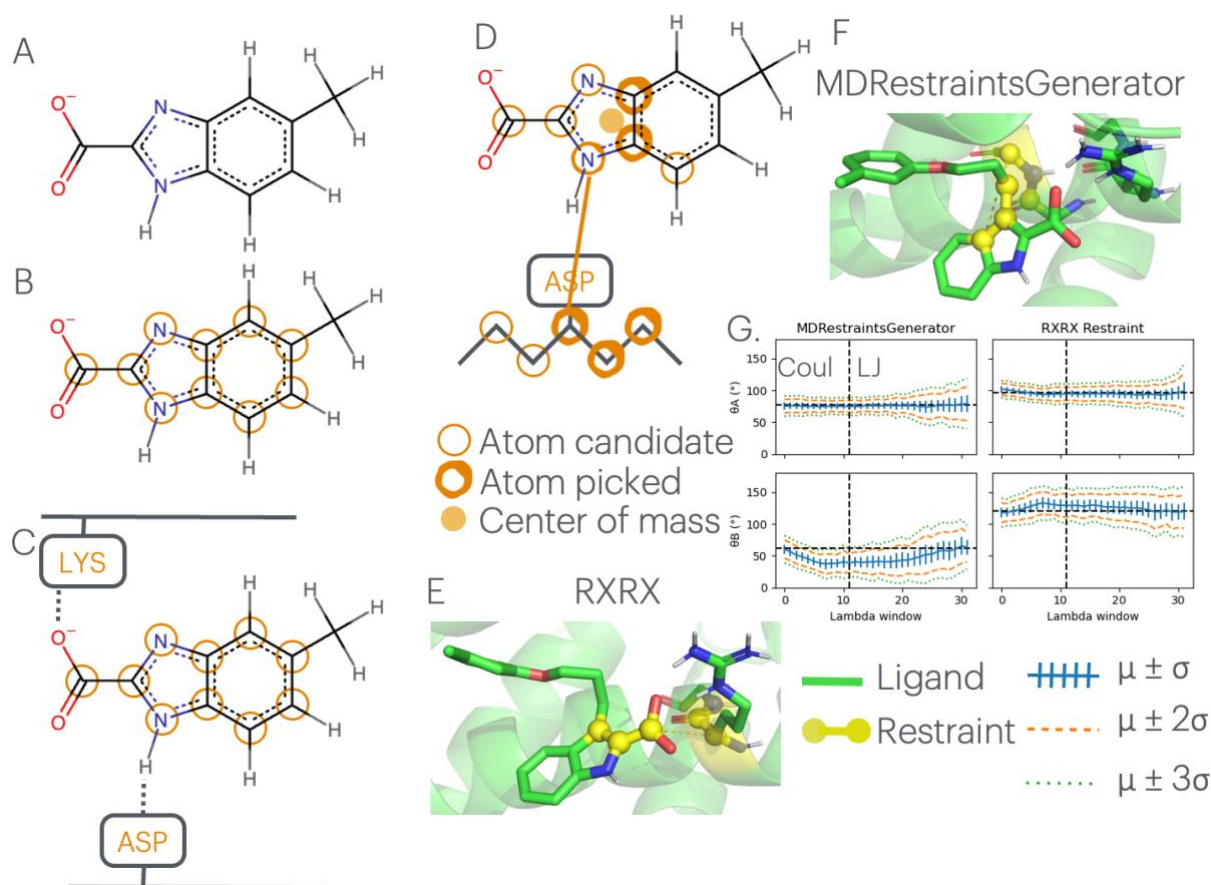


Figure 2. Process for deriving the atoms for an ABFE restraint. (A) Model ligand for illustrative purposes. (B) Select non-terminal atoms connected to at least two heavy atoms. (C) Mapping of hydrogen bonds (dotted line) between the protein and the ligand. (D) Selection of the hydrogen bond pair closest to the ligand's center of mass. (E) Six-atom restraint identified using the RRRX-style restraint search algorithm for the ligand lig_30 in the MCL1 set. (F) Six-atom restraint identified using MDRestraintsGenerator. (G) Mean μ and standard deviation σ of the two angles across all lambda windows (0-11 for charge annihilation and 11-32 for LJ annihilation). The RRRX restraints maintained a safe distance from 0 and 180 degrees across all windows, whereas within three standard deviations the MDRestraintsGenerator restraint for θ_A approached 0, leading to potential simulation instability.

To address this limitation, we developed a new restraint search algorithm inspired by previously reported methodologies²⁰. We illustrate the key steps for the hypothetical compound shown in Figure 2A. The non-hydrogen atoms connected to at least two heavy atoms are selected as candidate atoms for the ligand's restraint component (Figure 2B). We then enumerate all hydrogen bonds between the ligand and protein, mapping the ligand part of the hydrogen bond to the candidate atoms and the protein part of the hydrogen bond to the C α -carbon of the corresponding residue (Figure 2C). Only hydrogen bonds with an occupancy greater than 50% are selected, and all possible six-atom combinations (three atoms from the ligand and three from the protein backbone (C α , N, C)) that participate in these hydrogen bonds are considered. We filter these combinations further to retain only those where θ_A and θ_B fall within the range of 45 to 135 degrees. Among the remaining restraints, we select the ones in which the ligand atom is closest to the ligand's centre of mass (Figure 2D). For the

restraints that contain this ligand atom, we then select the one that gives the lowest score according to

$$score = \sigma(r)\sigma(\theta A)\sigma(\theta B)\sigma(\phi A)\sigma(\phi B)\sigma(\phi C)r^2(1 - \sin \theta A)(1 - \sin \theta B) \quad (1)$$

where the optimal restraint combination is chosen based on the standard deviation (σ) of the distance (r in nm), the angle values (θA and θB in radian), and the dihedral angle values (ϕA , ϕB and ϕC in radian). Equation 1, together with the earlier filtering steps, ensure that both angles are kept well away from 0 or 180 degrees during partial/full annihilation of the ligand's nonbonded interactions, and that the protein component is part of a well-ordered region (e.g., not a loop).

To compare the efficacy of this new restraint search algorithm with the previous approach, a protein-ligand system from the MCL1 dataset¹⁷ is shown which caused stability issues with the MDRestraintsGenerator. The restraints selected by the new approach are shown in Figure 2G, and a typical set of restraints obtained by MDRestraintsGenerator is shown in Figure 2F. While, in this example, both restraint selection algorithms pick ligand atoms near the centre of the compound, the distance restraint selected by the RXRX approach is aligned to a hydrogen bond which the ligand forms with the protein when it is fully interacting. With the two sets of restraints fully turned on (each pose restraint force constant was set to 10 kcal/mol), we then ran two series of simulations in which the nonbonded protein-ligand interactions were turned off, one for each set of restraints. Identical λ -windows (11 for the electrostatic interactions, λ spacing of 0.1; 21 for LJ, λ spacing of 0.05) were used in both cases.

During these annihilation simulations, we monitored the behaviour of θA and θB ; the results are shown in Figure 2G. The behaviour of θA (top plots) was inconspicuous in both cases, but, in the MDRestraintsGenerator case (left, bottom plot), the average θB value drops to 40 degrees at and after λ -window 10, i.e., when electrostatic interactions have been (mostly) switched off. In addition to the average values μ , the plots also indicate the $\mu \pm \sigma$ (blue), $\mu \pm 2\sigma$ (orange), and $\mu \pm 3\sigma$ ranges (green). Using the MDRestraintsGenerator selected restraint for θB , the $\mu - 3\sigma$ range approaches 0 degrees. While the plot might suggest that this is acceptable as an angle of 0 degrees is never reached, the following thought experiment will show how instabilities may arise in rare situations. Assuming that angle sampling follows a Gaussian distribution, the μ values and σ ranges shown for the θB /MDRestraintsGenerator case yield a probability of approximately 0.15% to encounter an angle of 0 degrees, which might cause a crash. To ensure statistical independence of the samples, we consider only samples (instantaneous angle values) drawn every 20 ps over a total simulation time of 5 ns (resulting in 250 independent samples). According to the binomial distribution, the probability that no sample (angle value) will be below $\mu - 3\sigma$ degrees is $(1 - 0.0015)^{250} = 69\%$. Consequently, there is a 31% chance that at least one sample (angle value) will be so close to 0 degrees that a crash will ensue. We stress that the observed crash rates using MDRestraintsGenerator were *much* lower than what the thought experiment suggests, but it is a reminder how extremely

unlikely individual events can lead to an undesirable event (crash) during a 5ns (or longer) simulation. Considering an application in a drug design project with dozens or hundreds of compounds, the probability of crashes increases even further.

Looking at the behaviour of θ_A with the RXRX restraints (right, bottom plot of Figure 2G), the angle range shifts somewhat from the target angle value towards 180 degrees, but the $\mu+3\sigma$ range remains at around 150 degrees. Since the full RXRX protocol also applies higher force constants to all angle restraints and turns them on at a later stage (see below), the probability of the critical angles of 0 and 180 degrees being approached becomes practically nil.

Approach to optimizing the λ -schedule

The Aldeghi protocol as described by Alibay et al.⁴ simulates each λ -window for 20ns. To reduce wall time, we wanted to lower the simulation length per window and use more λ -windows instead. The cloud resources on which Recursion's physics platform is implemented offers, e.g., instances with 48 and 64 cores, respectively. GROMACS,²³ the MD engine employed, can leverage all available CPU cores to accelerate simulations; however, it is essential that the number of CPUs in the instance are an integer multiple of the number of λ -windows. Thus, on these instances we can use 48 or 64 λ -windows combined with Hamiltonian replica exchange. In both the Aldeghi and the RXRX protocols, electrostatic and LJ interactions are switched off separately. Once one has decided how many λ -windows are used to turn off, e.g., the protein-ligand LJ interactions, the question of λ spacing arises. The Aldeghi protocol employs linear scaling (equidistant λ -windows) for electrostatics and a two-stage linear scaling for LJ interactions (wider equidistant spacing for $0 < \lambda < 0.5$, followed by a finer, equidistant spacing for $0.5 < \lambda < 1$).¹⁵ Given the cost of these types of calculations on large cloud instances, we explored whether other λ -schedules resulted in better performance (faster convergence, lower statistical error). Below we describe in detail the search for an optimal choice of 44 λ -windows for switching off LJ interactions, starting from an equidistant λ -schedule with 192 windows. Analogous optimizations were carried out for electrostatics (192 $\rightarrow$ 24 λ -windows) and LJ (192 $\rightarrow$ 25 λ -windows), results of which are briefly summarized in Figure S3. The 192 windows for the starting protocol correspond to the 192 vCPUs available on the AWS G5.48xlarge instance used for the reference simulations.

The λ -schedule optimization was based on the free leg, i.e., on simulations of the ligand in water, because in this case the convergence of the reference calculations can be guaranteed (see Results and Figure S4). The test systems are 16 ligands from the TYK2 benchmark. Each ligand was solvated in a cubic box with dimensions of 4 nm $\times$ 4 nm $\times$ 4 nm. Focusing on LJ annihilation, the starting state had zero partial charges with full LJ potential, and the end state had both zero partial charges and zero LJ protein-ligand interactions; however, the intramolecular LJ 1-4 interactions of the ligand were preserved.

The reference calculations for LJ annihilation used a linearly spaced grid of 192 λ -windows ($\lambda_i = i/191$, $i=0, \dots, 191$), with each window simulated for 5 ns. The timestep was set to 4 fs in

combination with hydrogen mass repartitioning. Replica exchange was attempted every 300 steps. Our goal was to find an optimal λ -schedule using only 44 λ -windows out of the total of 192 (Figure S5A). The end states, $\lambda=0$ and $\lambda=1$, were always included. The selection of λ -windows was achieved as follows (Figure S5B). Start with a random selection of 44 windows, always including $\lambda=0$ and $\lambda=1$, from the 192 available. Approximate a “gradient” for each λ -window, except for $\lambda=0$ and for $\lambda=1$, by evaluating the change in score upon changing index i for the particular λ_i by ± 1 . The three scoring schemes tested are described in detail below. The “gradient” was defined as the difference between the current score and the score of the two adjacent windows. For each window, the smaller of the two gradients was selected as the gradient for that window. The index to be optimized was selected randomly, with the probability of each index being chosen weighted according to the magnitude of its gradient. The selected window was then moved to the adjacent position that yielded a lower score. The optimization process was considered to have converged when all gradients became positive. This process is inherently stochastic and not fully reproducible. To ensure the λ -schedule with the lowest score is identified, the optimization process was repeated multiple times; see the below description of the three scores for the stop criteria.

Three scoring schemes were tested. The first score labelled as ‘OVERLAP’ was based on the MBAR overlap matrix (Figure S5C),²⁴ aimed to equalize overlap between adjacent windows, removing windows if overlap was too high and adding windows if it was too low. The score was the standard deviation of the logarithms of the overlaps between all adjacent windows. This approach assumes that the total error of the MBAR estimator is related to sample overlap, although this is a simplification.²⁵ This optimization for all repeats and 16 ligands took a few hours on a single CPU instance.

The second score, labelled as ‘BAR’, estimated the actual MBAR error by assuming total error as the sum of individual errors between adjacent windows, computed using the bootstrap error from the BAR estimator (Figure S5D). This approach simplistically assumes no overlap between states separated by more than one λ -window, which isn’t entirely accurate, as there is usually small but significant overlap beyond a single window that can affect the final result. This approach was accelerated by caching, i.e., error calculations between any two windows were precomputed from decorrelated samples and stored in a lookup table, speeding up the optimization. Like the OVERLAP score, this optimization completed within a few hours on a single CPU instance.

The third scoring method, labelled as ‘MBAR’, used the analytical error from the MBAR estimator (Figure S5E). This approach was computationally expensive because it requires a separate MBAR calculation for each error evaluation. Using the GPU JAX implementation of pymbar,²⁴ convergence from a random initial guess took four days for 44 windows.

For exhaustive searching, the overlap and BAR approaches were repeated with random initial guesses until no better solutions were found after 1000 consecutive attempts. For the much more expensive MBAR approach, 20 optimizations were run from random initial guesses,

with the best result selected. Because of the small number of iterations, the MBAR-optimized schedule was less smooth than the others. To improve smoothness, a polynomial fit was applied to the optimal schedule identified in this manner, and the fit function was used to obtain the optimal λ -schedule for the MBAR score.

Additional differences between the Aldegghi and RXXR protocols

The final RXXR protocol differs in three additional details from the Aldegghi one. First, in addition to the two selection algorithms for the restraints (see “Restraint search for the two ABFE protocols”), the force constants of the pose restraints are different in the two protocols. The Aldegghi protocol applies a uniform force constant of 10 kcal/mol for the distance, angle, and dihedral restraints, whereas the RXXR protocol uses a force constant of 1 kcal/mol/Å² for the distance and 80 kcal/mol/rad² for the angle and dihedral angle restraints. In particular the higher angle force constants were motivated by preventing potentially unstable θ values near 0 or 180 degrees. Moreover, in the Aldegghi protocol, the simulation frame closest to the mean distance, angle, and dihedral values averaged over a 1 ns equilibration run is used as the starting structure for the ABFE calculation. In contrast, the RXXR protocol, inspired by Chen et al.,⁹ selects the frame closest to the median value of all rotatable torsions of the ligand during the 1 ns simulation.

The second major difference concerns λ -scheduling of the bound state. In the Aldegghi protocol,²⁶ all six pose restraints are applied first, followed by the annihilation of electrostatic interactions, and then Lennard-Jones (LJ) annihilation, as can be seen from the detailed λ -schedules listed in Table S1. In contrast, the RXXR protocol starts with the annihilation of electrostatic interactions (21 λ -windows), followed by LJ annihilation (44 λ -windows), with the restraints being applied in two phases concurrently with the annihilation stages. Specifically, the three dihedral angle restraints are introduced during the first 32 λ -windows; then the distance and angle restraints are applied during the final 32 λ -windows (Table S1). For the detailed protocols, see “Production ABFE calculations” below; some additional details as well as the rationale for this choice will be presented in Results.

The third major difference lies in the alchemical end state. The Aldegghi protocol uses GROMACS default annihilation settings (``couple-intramol=True`, `couple-lambda0=vdw-q`, `couple-lambda1=None``), which retains Lennard-Jones (LJ) and Coulomb 1-4 interactions in the annihilated state. In contrast, the RXXR protocol sets the ligand’s alchemical dummy state by setting the ligand’s partial charges and LJ epsilon values to zero. Hence, the RXXR protocol avoids the GROMACS default settings which preserve the LJ 1-4 interactions. Additionally, rotatable torsions in the ligand are flattened using the SMARTS pattern “[!#1]~[!\$(***)&!D1]-!@[!\$(***)&!D1]~[!#1]”, which also includes amide bonds. To prevent flattening of the amide bonds, they are specifically excluded using the SMARTS pattern “[NX3][CX3](=[OX1])[#6]”. The force constants of selected torsions are scaled to zero with λ , but the multiplicity and phase parameters remain unchanged.

Production ABFE calculations

We applied our implementation of the Aldeghi protocol as well as the new RXXR protocol to four well established benchmark systems (TYK2, P38, JNK1, and CDK2) from the protein-ligand benchmark set¹⁷ in Chen et al.'s⁹ paper. These four systems all have neutral ligands, which avoids unrelated complications from changes in net charge. The protein and ligand coordinates were sourced from the protein-ligand benchmark repository.¹⁷ The protein was parameterized using the AMBER14SB²⁷ force field, while the ligands were parameterized with the OpenFF 2.2.0 force field.²⁸ System preparation was conducted with BioSimSpace,²⁹ where the bound leg employed a dodecahedral box with buffer length of 0.5 nm, and the free leg utilized a cubic 4 nm × 4 nm × 4 nm box. Hydrogen mass repartitioning was applied, scaling hydrogen masses to 4 atomic mass units (a.m.u.). A 1 nm cutoff was used for LJ interactions and Particle Mesh Ewald (PME).³⁰ Replica exchange was performed every 300 steps.

For the bound leg, the system underwent energy minimization followed by a 10-step equilibration protocol³¹ using GROMACS 2024.2.²³ After equilibration, the system was simulated for 1 ns with a 4 fs timestep to generate frames for restraint searches, as already described. All λ -windows were then energy-minimized for 5000 steps, followed by equilibration at 298K for 40 ps.

The Aldeghi protocol's⁴ recent implementation used 20 ns simulations per λ -window in combination with 24 λ -windows for the free and 32 λ -windows for the bound leg of the thermodynamic cycle (Table S1). To limit the computational cost in high-throughput calculations, the RXXR protocol employs twice the number of λ -windows (48 for the free, 64 for the bound leg), but only 5 ns simulations per λ -window (see next paragraph). To maintain an equivalent total sampling time, our implementation of the Aldeghi protocol, therefore, uses only 10 ns simulations per λ -window. For the bound leg in the Aldeghi protocol, first empirical geometric scaling (Table S1) is applied to all six protein-ligand pose restraints; i.e., all restraints are fully active when the charge annihilation starts. Charges are linearly scaled to zero in 6 windows, followed by 16 windows of two-stage linear LJ annihilation, with a λ spacing of 0.1 for $\lambda < 0.5$ and 0.05 for $\lambda \geq 0.5$. For the free leg, charge is linearly scaled to zero in 9 windows, followed by LJ annihilation in 16 windows.

Under the RXXR protocol, production runs were carried out for 5 ns per λ -window (in total, 48 λ -windows for the free, 64 λ -windows for the bound leg). In the bound leg of the RXXR protocol, the dihedral components of the protein-ligand pose restraint is turned on according to a geometrical progression during the first half (32 λ -windows), starting at 0 for the first λ -windows and 0.01 for the second λ -window, while the bond and angle components are only turned on analogously during the second half. Charge is linearly annihilated during the first 21 λ -windows with a λ spacing of 0.05, and LJ-interactions are annihilated in 44 windows, following the optimized λ -schedule obtained from the MBAR score (see Results for the performance of the three scoring methods tested). The internal dihedral angle potentials are flattened in parallel with applying the distance and angle components of the protein-ligand

pose restraints, using the geometric progression of λ -values as described above. For the free leg, charge is linearly annihilated in the first half, with van der Waals forces annihilated in 25 windows (second half). The internal dihedral angle potentials of the ligand are flattened with linear scaling over all λ -windows. All details of the λ -schedules are listed in Table S1; they are also depicted graphically in Figure 3C.

The resulting free energy data were analysed using alchemlyb³². For each protein-ligand system, the full ABFE calculation was repeated three times with both protocols; these repeats were used for error estimation when comparing to the experimental data.

Results and discussion

Evaluating λ -schedules

As discussed in Methods, the optimization depends on the convergence of the reference calculations (the 192 windows protocol). This is the reason why the optimization procedure was carried out for the ligand(s) in water. Converged simulations of solute-solvent systems are doable with today's means, but it is still impossible to guarantee the convergence of protein-ligand simulations. The convergence of the 192 windows reference calculations for the ligands in water is illustrated in Figure S4. It shows the convergence of the free energy difference for turning off LJ interactions as a function of simulation time for each of the 16 ligands. Convergence is plotted in the forward (blue) and backward (red) direction. Note the small scale on the y-axis; even after only 10% of the simulation time per λ -window, the largest deviation from the final result is 0.1 kcal/mol.

Having established the convergence of the reference data, we plot in Figure 3A the optimized 44 λ -windows schedules obtained with the three scoring approaches, OVERLAP (orange), BAR (green), and MBAR (blue). We also include a strictly linear schedule (red), and an Aldeghi-like two-step linear schedule (purple). Here, the λ spacing between $\lambda=0.0$ and $\lambda=0.5$ is twice as large as the spacing between $\lambda=0.5$ and $\lambda=1.0$, thereby allocating more λ -windows to the second half of the transformation. One sees in Figure 3A that the λ -schedule derived using the BAR score closely resembles that from the MBAR score. Both are quite similar to linear scheduling, whereas the optimal schedule found using the OVERLAP score deviates noticeably from the linear path.

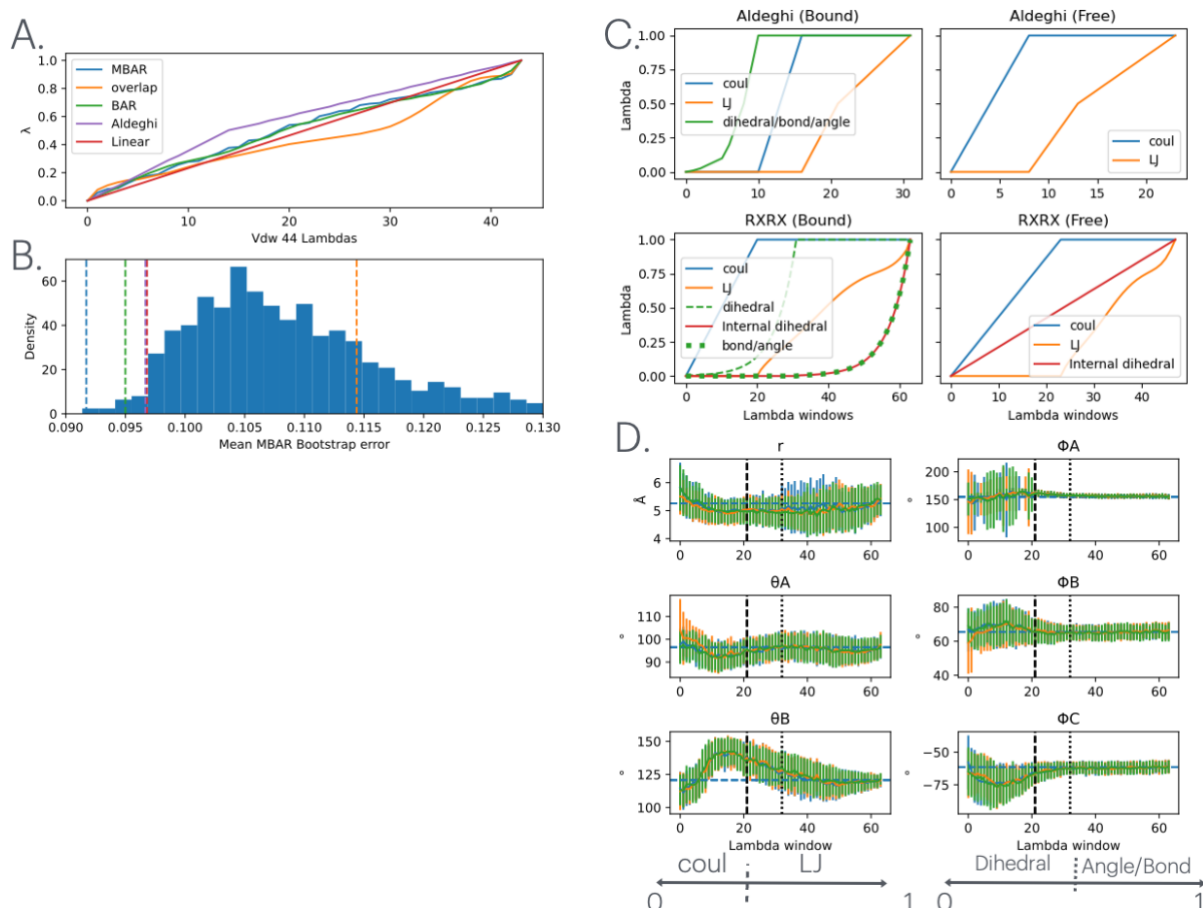


Figure 3. (A) Optimal λ -schedules derived from the OVERLAP score (orange), the MBAR analytic error score (blue), the score based on error propagation via bootstrap BAR error (green), and an Aldegghi-like scheme (purple) and linear (red) for LJ annihilation with 44 windows. (B) Distribution of the MBAR bootstrap error for randomly chosen lambda schedules. The results obtained with the three optimized protocols and the Aldegghi scheme are indicated; the same colour code as in (A) is used. (C) Comparison of the λ -scheduling of the Aldegghi protocol (top) and the RXRX protocol (bottom). In the bound leg, charge is linearly annihilated during the first 21 windows, followed by nonlinear LJ annihilation using the MBAR-optimized schedule. Torsional restraints are added via geometric scaling during the first half of the simulation, while bond and angle restraints are added in the second half using the same scaling. The internal torsion potential is removed in the same manner as bond and angle restraints are applied. For the free leg, charge is annihilated in the first half, and LJ interactions are annihilated non-linearly (see Figure S3). The internal torsions are annihilated linearly during all λ -windows. (D) Six degrees of freedom for the restraint under the RXRX lambda scheduling for ligand lig_30 in the MCL1 set. The different colours represent three separate repeats; each bar represents $\mu \pm \sigma$. Angles are most mobile at the end of charge annihilation (vertical, black dashed line), where they are, therefore, most prone to approach 0 or 180 degrees. In the RXRX protocol, this stage occurs before the angle restraints are applied (indicated by the vertical, black dotted line).

The various λ -schedules were evaluated as follows: As the null hypothesis, we generated 1,000 random λ -schedules from the complete set of 192 λ -windows for the 16 TYK2 ligands in solution. For each of these random schedules, we computed the free energy difference and bootstrap error using MBAR²⁴. The resulting distribution of error estimates is displayed as a histogram in Figure 3B. The error estimates for the OVERLAP, BAR, MBAR, Aldeghi-like and linear schedules (same colour scheme as in Figure 3A) are overlaid on the histogram of the random error distribution. The worst performer is the OVERLAP score derived schedule; all others perform quite well, i.e., have a smaller bootstrap error estimate than (most of) the random schedules. The MBAR score derived schedule performs best. Figure S3 shows the analogous analysis for the optimization of a 24 λ -windows electrostatics and a 25 λ -windows LJ annihilation schedule. Since in both cases, the MBAR score derived schedule also performed best (i.e., had the lowest bootstrap error), we expect that choosing the optimized schedule based on the MBAR score for LJ annihilation will reduce the statistical error of ABFE calculations. That the MBAR score led to the best ranking aligns with expectations, as the analytical error from MBAR can be expected to be closely correlated with its bootstrap error.

Comparison of RXRX and Aldeghi Protocols for λ -scheduling and Restraints

Figure 3C depicts the overall λ -schedules (turning off nonbonded interactions, applying restraints, flattening of intra-ligand torsional barriers (RXRX protocol only)) as used in our production calculations with the Aldeghi and RXRX protocols for the free and bound leg of the ABFE thermodynamic cycle. These data are also tabulated in Table S1. The plots highlight the different number of λ -windows and the differences in scheduling of the two protocols, which were already described in Methods. Here, we focus on the effects of these differences on the restraint degrees of freedom, which were the motivation for the reordered scheduling of the RXRX protocol. Figure 3D shows the average values $\pm$ one standard deviation of the six pose restraint degrees of freedom for the 64 λ -windows in the RXRX protocol. The data (three repeats shown in different colours) were obtained for ligand *lig_30* in the MCL1 set, for which related data were already shown in Figure 2G. Note, however that Figures 2G and 4B differ in the number of λ -windows and the force constants of the restraints. One sees that the two restraint angles, particularly θ_B , deviate most from their target values (dashed horizontal lines) when the electrostatic interactions are close to or at zero (λ -windows 10-21, the completion of the annihilation of electrostatic interactions is indicated by the vertical, dashed black line). Interestingly, angle values return to their target values once we start to turn off LJ interactions, i.e., after λ -window 21. The RXRX approach to select restraints and the higher force constants in the RXRX protocol should by themselves prevent the angles from approaching 0 or 180 degrees. However, studying plots like the ones shown in Figure 3D motivated us to introduce the angle restraints only *after* electrostatic interactions have been completely removed. The introduction of the angle (and the distance) restraints is indicated by the dashed, vertical black line. Because the angles fluctuate most when there are no angle restraints present, a crash cannot occur. We further see from Figure 3D that the distance degree of freedom remains well-behaved over all λ -windows, even

though it is applied only after λ -window 32 and even though its force constant is low. Similarly, the three dihedral degrees oscillate about their target values while the restraints are turned on during the first 32 λ -windows, and then remain in a narrow range once the restraints are fully active. While Figure 3D only shows data for a single protein-ligand system, aggregate data obtained for all benchmark compounds, which we show below, confirm the observations and conclusions drawn from Figure 3D and rationalise the decision to apply the three dihedral restraints first and before the distance and angle restraints.

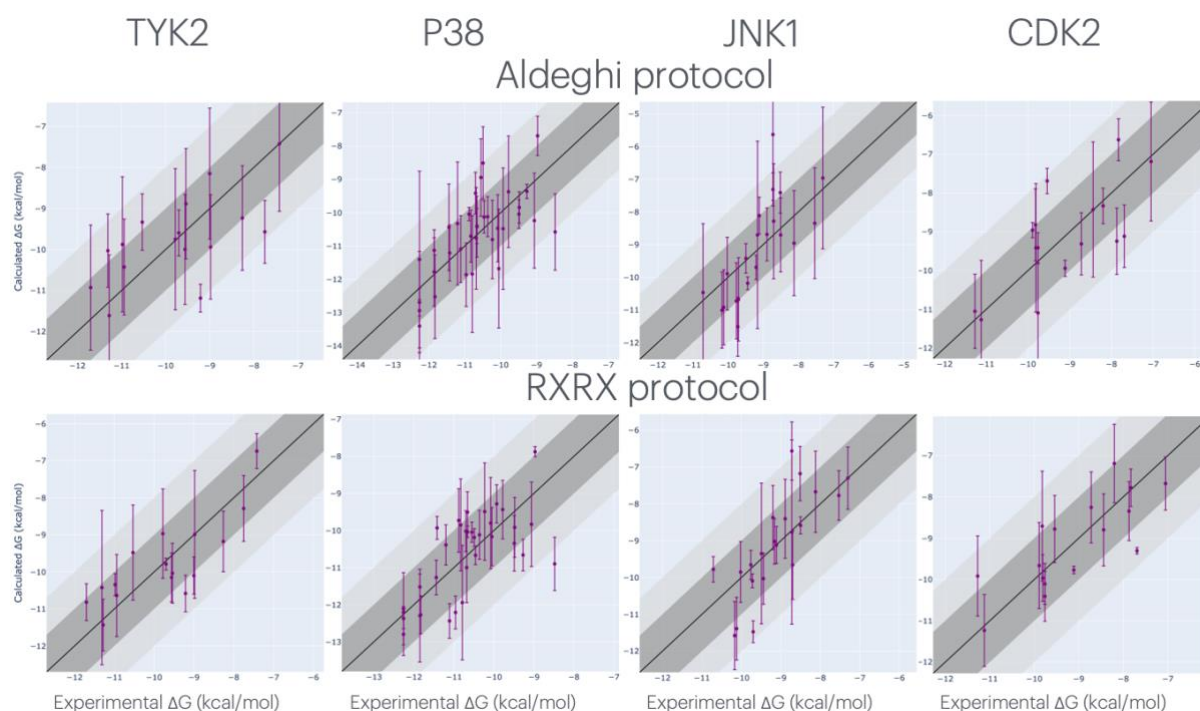


Figure 4. Benchmark comparison of the Aldeghi and RXXR protocols across four protein systems (TYK2, P38, JNK1, CDK2). The x-axis shows the experimental reference results and the y-axis the computed binding free energies. Deviations within 1 and 2 kcal/mol are highlighted by the dark gray and light gray areas. Error bars represent the standard deviation across three repeats. The RXXR protocol consistently results in lower standard deviation between repeats compared to the Aldeghi protocol.

Benchmark of the new protocol

As described in Methods, protocol RXXR (new restraint search method, optimized/modified λ -scheduling) was benchmarked against four neutral systems (TYK2, P38, JNK1, and CDK2) from the protein-ligand benchmark set¹⁷ in Chen et al.'s⁹ paper. All ABFE simulations were also carried out with and compared to our implementation of the Aldeghi protocol (MDRestrainsGenerator to generate restraints, simulation time of 10 ns per λ -window to match our total simulation time for a fair comparison, λ -scheduling as shown in Figure 3C, top panels). Three repeats were conducted for each ligand. The results are summarized in Table 1 and Figures 4 and 5A-B. The statistical descriptors of the results compared to the experimental binding affinities for the RXXR protocol are with one exception (Kendall's τ for P38) better than for the Aldeghi results (Table 1). E.g., the root-mean-square errors (RMSE) drop between 0.05 (P38) and 0.23 (TYK2) kcal/mol using the RXXR instead of the

Aldeghi protocol. The RMSE of the two protocols used here, as well as those reported by Schrodinger for their RBFE and ABFE⁹ protocols are also depicted in Figure 5B. The RXXR RMSE (green bars) is always lower than that obtained with the Aldeghi protocol (dark red bars) and compares favourably with the Schrodinger protocols (blue and golden bars).

A particular challenge with ABFE calculations is the significant variance between repeats, which is normally larger than in RBFE.⁹ The plots of individual results for each of the ligands in the four systems (Figure 4) indicate that the standard deviation across the three repeats is smaller for RXXR than for Aldeghi. A more concise summary of this finding is shown in Figure 5A. There, one clearly sees that RXXR protocol consistently led to smaller statistical errors across the three repeats compared to Aldeghi.⁹

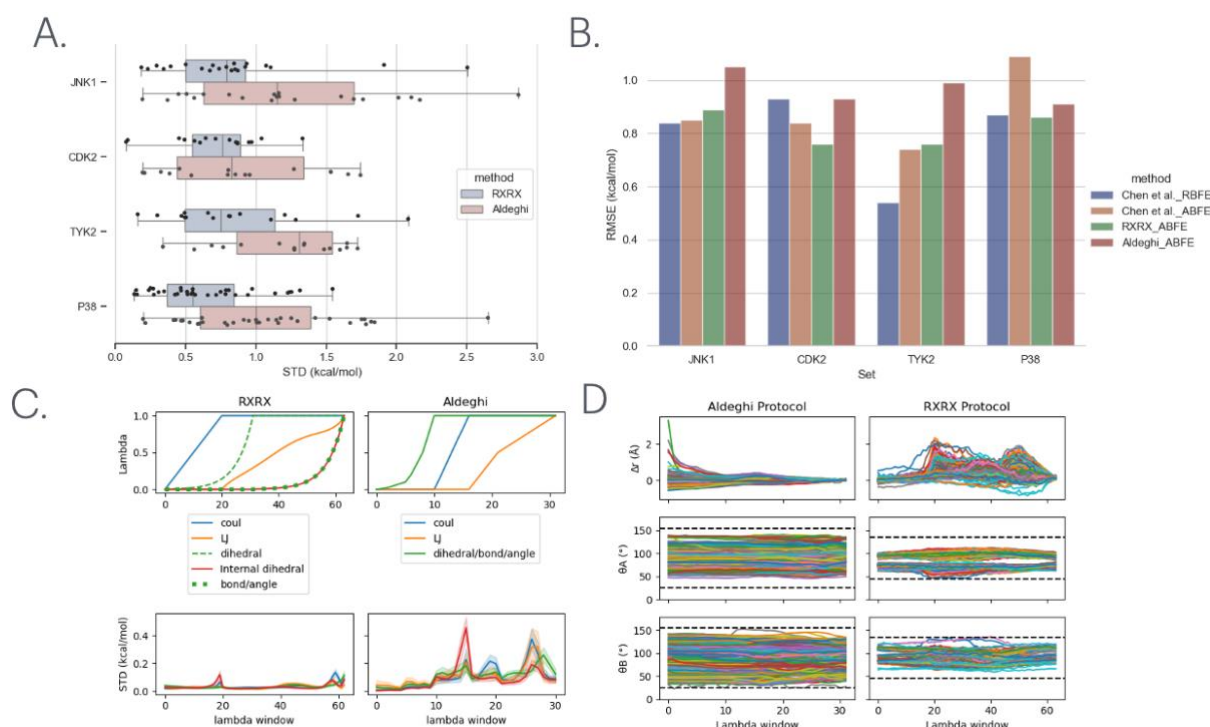


Figure 5. A. Comparison of the standard deviation distribution across three repeats for all compounds using the RXXR protocol versus the Aldeghi protocol. B. RMSE of the computed ΔG_{bind} with respect to experimental values for RXXR, Aldeghi and Chen et al. RBFE/ABFE for the four benchmark systems. C. Standard deviation of $\Delta G_{i,i+1}$ between adjacent lambda windows are consistently lower with the RXXR protocol. D. Deviation from the target distance restraint value Δr and angle restraint values θA and θB averaged over all repeats for each ligand using the RXXR and Aldeghi restraint schemes.

Figures 5C and D provide additional insights into the differences between the two protocols, as well as why RXXR lowered standard deviations across repeats. For Figure 5C, we calculated free energy differences $\Delta G_{i,i+1}$ for all pairs of neighbouring λ -windows and their standard deviations over the three repeats. The standard deviations $\sigma(\Delta G_{i,i+1})$ as a function of λ -window are plotted in the bottom two panels for the RXXR (left) and Aldeghi protocols (right). The two top panels show again the order in which interactions/restraints are turned on/off as a function of λ -windows. Data for all ligands of the four benchmark systems were used. First, the plots show that overall the Aldeghi protocol leads to higher standard

deviations $\sigma(\Delta G_{i,i+1})$, in line with the bigger spread of results across repeats (Figure 5A). Second, one can discern immediately for which λ -windows $\sigma(\Delta G_{i,i+1})$ is high, i.e., where most of the statistical error arises. In the Aldeghi protocol, a peak in $\sigma(\Delta G_{i,i+1})$ occurs at the end of charge annihilation (λ -windows 15-18). In the new protocol, slightly higher $\sigma(\Delta G_{i,i+1})$ values at the end of charge annihilation (around λ -window 20) can also be observed but with substantially reduced magnitude. This difference can be attributed to the fully activated restraint during charge annihilation in the Aldeghi protocol, which, while restricting the accessible configuration space, also increases the energy barrier between different configurations. This results in different repeats sampling distinct configuration spaces with slow interconversion. Conversely, the RXRX protocol only activates the dihedral restraint during charge annihilation, providing sufficient restriction to the configuration space while still allowing reasonable interconversion between states. A second peak is observed in the Aldeghi protocol near the end of LJ-annihilation. Again, a similar peak can be discerned in the RXRX protocol, but it is flattened in the new protocol due to the λ -scheduling optimization, which identifies this problematic region and introduces more λ -windows with a gentler slope in this area.

In Figure 5D we plot Δr , the deviation of the distance restraint degree of freedom from its target value, and the values of the two angle degrees of freedom for all ligands/repeats as observed in the Aldeghi and RXRX protocols. In the RXRX protocol, the bond restraint is gradually introduced only during the second half of the simulation. Further, its force constant is low (1 kcal/mol/Å²), raising concerns that the ligand might unbind after/during charge annihilation. As one sees (top plot, left column in Figure 5D), the maximum deviation Δr observed is 2 Å. This is larger than in the Aldeghi protocol, where the distance restraint is introduced earlier and has a higher force constant but seems to have no adverse effects. By contrast, the two angle degrees of freedom, θ_A and θ_B remain in a much narrower range when using the RXRX protocol. On the one hand, their target values are chosen from a narrower range, $45 < \theta < 135$ degrees for RXRX vs. $25 < \theta < 155$ degrees for Aldeghi. On the other hand, the RXRX force constants for the angle restraints are 8 times higher than those of Aldeghi (80 vs. 10 kcal/mol/radian²). Most importantly, however, even though the angle restraints in RXRX are introduced only after λ -window 32, the angle values observed remain within the $45 < \theta < 135$ degrees range for all ligands studied here, safely away from the critical areas near 0 and 180-degrees. The aggregate data for all ligands of the four benchmark sets and all repeats shown in Figures 5C and D support the choices made concerning restraint selection and handling in protocol RXRX.

Conclusions

We described several modifications to an established ABFE workflow which is widely used,^{4,8,15} to enhance the stability of the simulations and to improve the reproducibility of results over repeats. The final RXRX protocol is in production use at Recursion. Our changes to the original Aldeghi protocol included (1) a change of the automated restraint selection

algorithm (carefully optimizing restraint setup to align with key protein-ligand interactions), (2) searching for an optimized λ -schedule to annihilate LJ interactions, and (3) reordering the annihilation of nonbonded protein-ligand interactions and turning on the pose restraints. In addition, intramolecular torsions of the ligand are flattened at the non-interacting end states. ABFE simulations using the RXXR protocol are considerably more stable; we observed no crashes applying it to the four protein-ligand benchmark systems, as opposed to a low single digit failure rate using the Aldeghi protocol. Figures 4 and 5A demonstrate that the variation of computed binding free energies ΔG_{bind} is significantly lower using the RXXR compared to the Aldeghi protocol; i.e., the reproducibility of ABFE calculations over repeats is improved. Furthermore, compared to the binding affinities (Table 1, Figure 5B) we obtained lower RMSE and improved correlation, with a single exception. Since the identical force field and the same MD program were used in all simulations, these improvements result from the changes in protocol.

It's difficult to attribute the improvements to a specific modification. Our data suggest, e.g., that the stability of ABFE simulations with restraints obtained from MDRestraintsGenerator could be improved by adopting a tighter angle filter, i.e., allow only angle restraints with $45^\circ < \theta < 135^\circ$ degrees, rather than the current threshold of $25^\circ < \theta < 155^\circ$ degrees. However, the higher force constants for the angle restraints, as well as introducing them only *after* electrostatic interactions have been switched off, also contribute to the higher stability of the RXXR protocol

Some details of the RXXR protocol are tied to the hardware/software stack being used at Recursion, e.g., the number of λ -windows used. However, all optimizations / modifications are quite general and can be adapted to one's use case. E.g., the LJ annihilation protocol can be repeated to search for an optimal protocol using 31 λ -windows starting from a 128 λ -windows reference simulation. More importantly, plots like Figure 5C make it easy to discern hotspots where statistical uncertainty is high. Similarly, plots like Figure 3D or 5D validate the protocol choices made; analyses like these make it straightforward to carry out protocol optimizations to adapt the ideas presented here to a different (software/hardware) environment.

Compared to RBFE, which are today used almost routinely,⁷ ABFE are more versatile and can more easily handle, e.g., dissimilar scaffolds. However, they tend to be more costly and when applied to tasks that can also be handled by RBFE, the precision of the results may be lower.⁹ The protocol presented here, already in real-world use at Recursion, demonstrates that there are stability and statistical gains to be had by re-arranging and optimizing restraints and λ -windows. Considering that the benefits of ABFE outweigh the costs, we expect to see more applications of ABFE in active drug design projects in the future.

626 Table 1. Benchmark results of the ABFE for the RXRX and Aldegghi protocol

	RMSE (kcal/mol)	MUE (kcal/mol)	Pearson's r^2	Kendall's τ
RXRX Protocol				
TYK2	0.76 [0.59 0.91]	0.66 [0.47 0.84]	0.64 [0.28 0.84]	0.74 [0.32 0.93]
P38	0.86 [0.66 1.09]	0.69 [0.53 0.87]	0.48 [0.23 0.72]	0.46 [0.23 0.65]
JNK1	0.89 [0.59 1.16]	0.64 [0.40 0.92]	0.60 [0.42 0.79]	0.61 [0.41 0.78]
CDK2	0.76 [0.52 0.98]	0.62 [0.42 0.85]	0.60 [0.31 0.83]	0.49 [0.21 0.73]
Aldegghi Protocol				
TYK2	0.99 [0.70 1.27]	0.81 [0.55 1.06]	0.41 [0.05 0.73]	0.48 [0.16 0.76]
P38	0.91 [0.72 1.12]	0.73 [0.55 0.92]	0.46 [0.24 0.69]	0.50 [0.31 0.69]
JNK1	1.05 [0.66 1.47]	0.81 [0.55 1.09]	0.55 [0.36 0.79]	0.61 [0.37 0.79]
CDK2	0.93 [0.65 1.16]	0.75 [0.49 1.02]	0.51 [0.10 0.78]	0.48 [0.18 0.75]

627 The calculated dG is the mean of the three repeats. The [] indicates the 95% confidence
 628 interval.

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653 ABBREVIATIONS

654 CCR2, CC chemokine receptor 2; CCL2, CC chemokine ligand 2; CCR5, CC chemokine
655 receptor 5; TLC, thin layer chromatography.

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